

Class 10

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The PDB Database

The [Protein Data Bank \(PDB\)](#) is the main repository of biomolecular structure data. Let's see what is in it:

```
stats <- read.csv("pdb_stats.csv", row.names = 1)
stats
```

	X.ray	EM	NMR	Integrative	Multiple.methods	Neutron
Protein (only)	178795	21825	12773	343	226	84
Protein/Oligosaccharide	10363	3564	34	8	11	1
Protein/NA	9106	6335	287	24	7	0
Nucleic acid (only)	3132	221	1566	3	15	3
Other	175	25	33	4	0	0
Oligosaccharide (only)	11	0	6	0	1	0
	Other	Total				
Protein (only)	32	214078				
Protein/Oligosaccharide	0	13981				
Protein/NA	0	15759				
Nucleic acid (only)	1	4941				
Other	0	237				
Oligosaccharide (only)	4	22				

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
n.sums <- colSums(stats)
n <- n.sums/n.sums["Total"]
round(n, digits = 2)
```

	X.ray	EM	NMR	Integrative
	0.81	0.13	0.06	0.00
Multiple.methods		Neutron	Other	Total
	0.00	0.00	0.00	1.00

X.ray is about 81% and EM is about 13%.

What is the total number of entries in PDB?

```
n.sums["Total"]
```

```
Total
249018
```

Q2: What proportion of structures in the PDB are protein?

```
rowSums(stats)
```

	Protein (only)	Protein/Oligosaccharide		Protein/NA
	428156	27962		31518
Nucleic acid (only)		Other	Oligosaccharide (only)	
	9882	474		44

Using Molstar

We can use the main [Molstar viewer online](#):

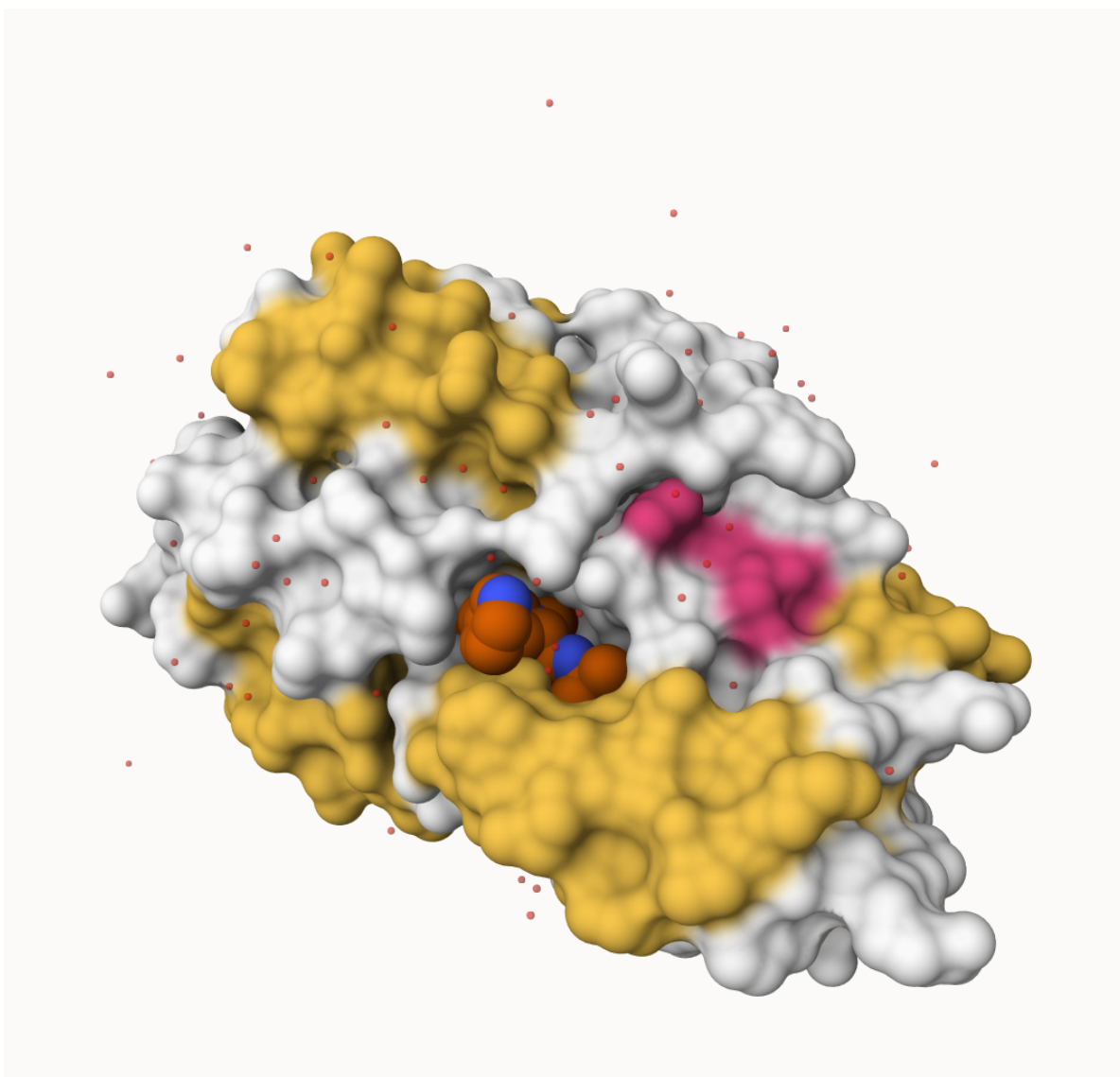
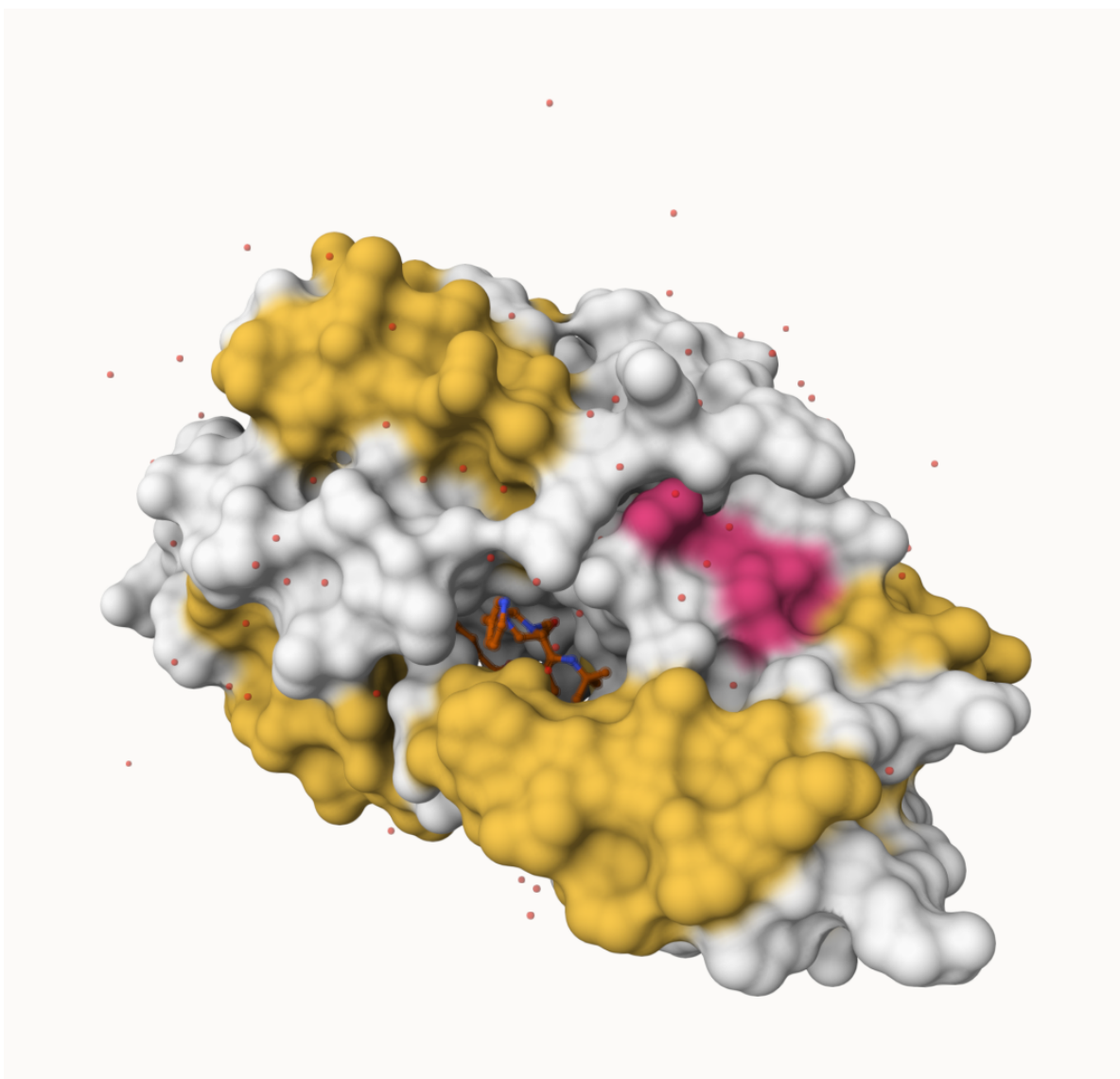


Figure 1: First view of HIV-Pr dimer with bound inhibitor

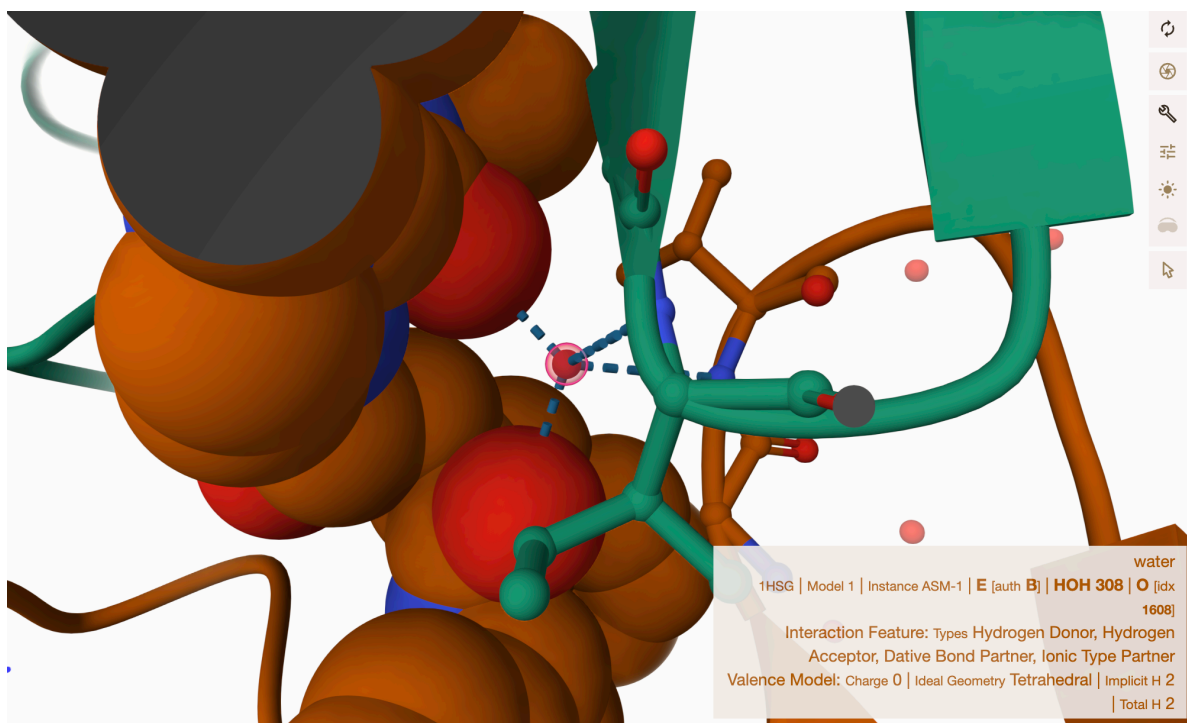
Q. Generate and insert an image of the HIV-Pr cartoon colored by secondary structure, showing the inhibitor (ligand) in ball and stick.



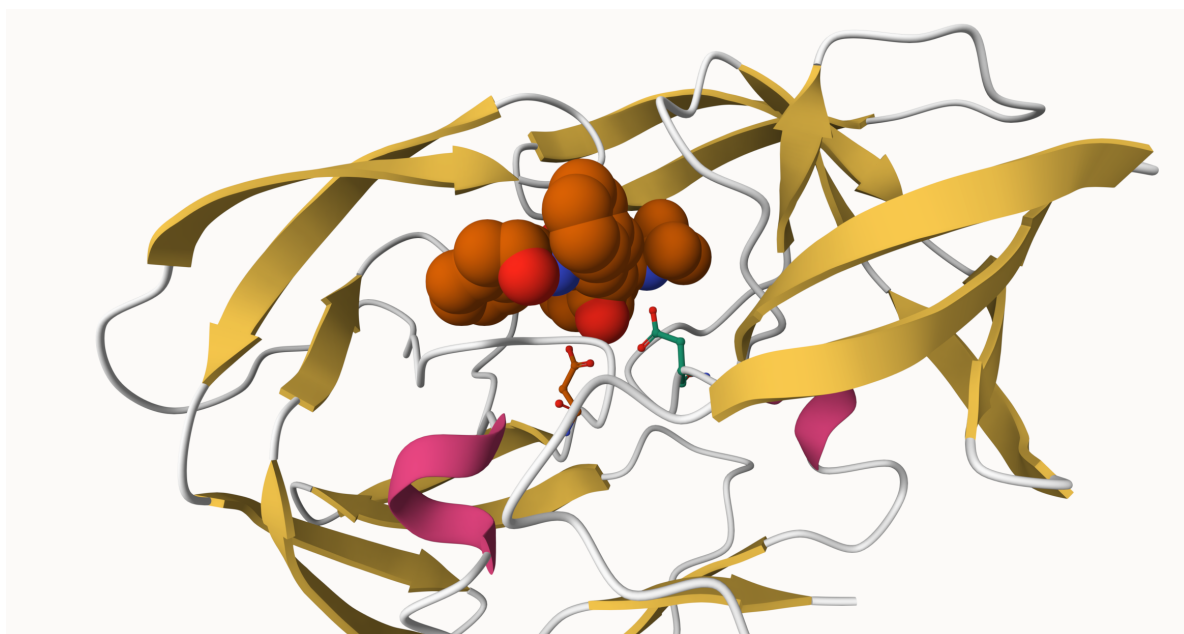
Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

They are condensed so we can analyze the structure of the protein and ligand better.

Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have



Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.



Introduction to Bio3D in R

Bio3D is an R package for structural bioinformatics. Features include the ability to read, write and analyze bio-molecular structure, sequence and dynamic trajectory data.

```
library(bio3d)
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
  Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

```
Protein sequence:
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

198

Q8: Name one of the two non-protein residues?

HOH

Q9: How many protein chains are in this structure?

```
attributes(pdb)
```

```
$names
```

```
[1] "atom" "xyz" "seqres" "helix" "sheet" "calpha" "remark" "call"
```

```
$class
```

```
[1] "pdb" "sse"
```

```
head(pdb$atom)
```

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

Let's try out the new **bio3dview** package that is not yet on CRAN. We can use the **remotes** package to install any R package from Github.

```
library(bio3dview)
library(NGLViewerR)

#view.pdb(pdb, backgroundColor = "pink") |>
# setSpin()
```

```
library(bio3dview)
library(NGLViewerR)

#view.pdb(pdb) |>
```

```
# setSpin()
#sele <- atom.select(pdb, resno=25)
# view.pdb(pdb, backgroundColor = "pink", cols=c("navy","teal"),
#         highlight = sele,
#         highlight.style = "spacefill") |>
# setRock()
```

Prediction of Protein flexibility

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
 PDB has ALT records, taking A only, rm.alt=TRUE

```
adk
```

```
Call: read.pdb(file = "6s36")
```

```
Total Models#: 1
```

```
Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)
```

```
Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 244 (residues: 244)
```

```
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

```
Protein sequence:
```

```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV
DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
VGRRVHAPSGRVYHVKFNPVKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

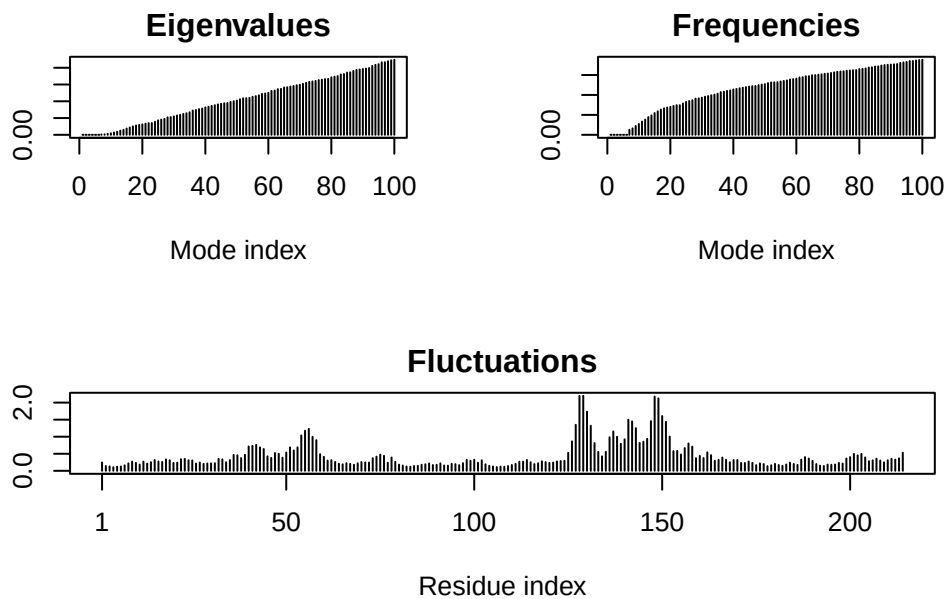
```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```



```
m <- nma(adk)
```

```
Building Hessian...      Done in 0.014 seconds.  
Diagonalizing Hessian... Done in 0.06 seconds.
```

```
plot(m)
```



Write out our results as a trajectory movie:

```
mktrj(m, file="adk_m7.pdb")
```

```
#view.nma(m, pdb=adk)
```

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

TRUE

```
library(bio3d)  
aa <- get.seq("lake_A")
```

Warning in get.seq("lake_A"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

aa

```

      1      .      .      .      .      .      .      60
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLV
      1      .      .      .      .      .      .      60

      61      .      .      .      .      .      .      120
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      .      120

     121      .      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
     121      .      .      .      .      .      .      180

     181      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
     181      .      .      .      214
```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

```
+ attr: id, ali, call
```

Q10. Which of the packages above is found only on BioConductor and not CRAN?

msa

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

214